

BDM Field Intelligence & Territory Network Audit Engine

Retail Churn Prevention & Channel-Adaptive Field Operations for Tamil Nadu

Forward Deployment Engineer

12 Regional Territories (Madurai, Chennai, Coimbatore, etc.)	820+ Multi-Tier Retail Counters Managed	>45 Days Automated Dormancy Flagging	1-Command Multi-Stage Docker Container Deployment
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Executive Summary

General Trade and specialist Apple retail counters across Tamil Nadu routinely go quiet for weeks before a distributor notices the revenue impact, because dormancy signals sit buried in static spreadsheets rather than surfacing to the field team in real time. Meanwhile, a single one-size-fits-all store checklist cannot serve both a General Trade counter, which needs credit and margin checks, and an Apple Premium Reseller, which must be audited against Apple's Experience Table standards. FieldBeat TN closes both gaps with a full-stack, tier-adaptive field intelligence platform: it automatically flags at-risk outlets, drives Business Development Managers through channel-specific 5-point audits, and issues cryptographically verifiable digital receipts for every visit commitment. The result is a single, deployable system that replaces fragmented spreadsheets and unverifiable paper trails with a prioritized, auditable field operations workflow.

The Business Problem & Domain Reality

Problem Statement & Territory Realities

Three Core Failure Points

- **Quiet Churn Blindspots:** Outlets inactive for **>45 days** get buried in static spreadsheets until distributor revenue is lost.
- **Checklist Mismatch:** General Trade stores need credit and margin checks, while APRs require Apple Experience Table compliance. A single generic checklist fails both.
- **Unverified Verbal Commitments:** Paper or verbal order commitments lack cryptographic proof and real-time ledger persistence.

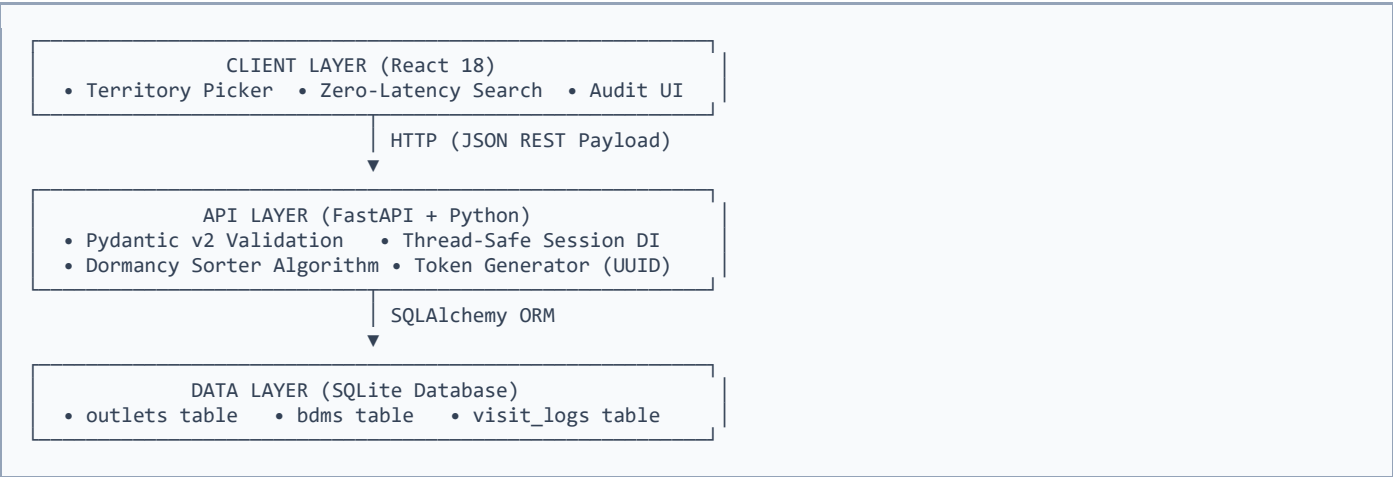
OBJECTIVE

An automated, tier-adaptive assistant prioritizing at-risk stores and generating tamper-evident digital receipts.

System Architecture & Data Flow

Full-Stack System Architecture

High-Level Architecture Overview



1. Layer-by-Layer Technical Breakdown

Frontend Architecture (React 18 + Vite + Tailwind CSS):

- **State Management:** Localized reactive hooks (`useState`, `useMemo`) managing active territory filtering, multi-attribute text search, and 5-point checklist completion state.
- **Data Visualization:** Lightweight SVG sparkline components rendering 6-month historical unit movement volume curves.
- **Adaptive Forms:** Dynamic form rendering engine that swaps audit criteria automatically based on the selected counter's `channel_tier` (General Trade, Mobile Specialist, APR).

Backend Architecture (FastAPI + Python 3.11):

- **REST Engine:** Async route handlers running under Uvicorn ASGI with CORS middleware.
- **Data Validation:** Strict type-enforced Pydantic v2 schemas (`VisitSubmitRequest`, `VisitSubmitResponse`, `OutletResponse`).
- **Session Management:** Per-request isolated database session injection via `Depends(get_db)` pattern.

Data Layer (SQLite + SQLAlchemy ORM):

- Relational schema comprising three core entities: `BDM` (territory leads), `Outlet` (retail stores, tiers, billing history), and `VisitLog` (audit records and cryptographic receipts).
- Concurrency safety enabled via `connect_args={"check_same_thread": False}`.

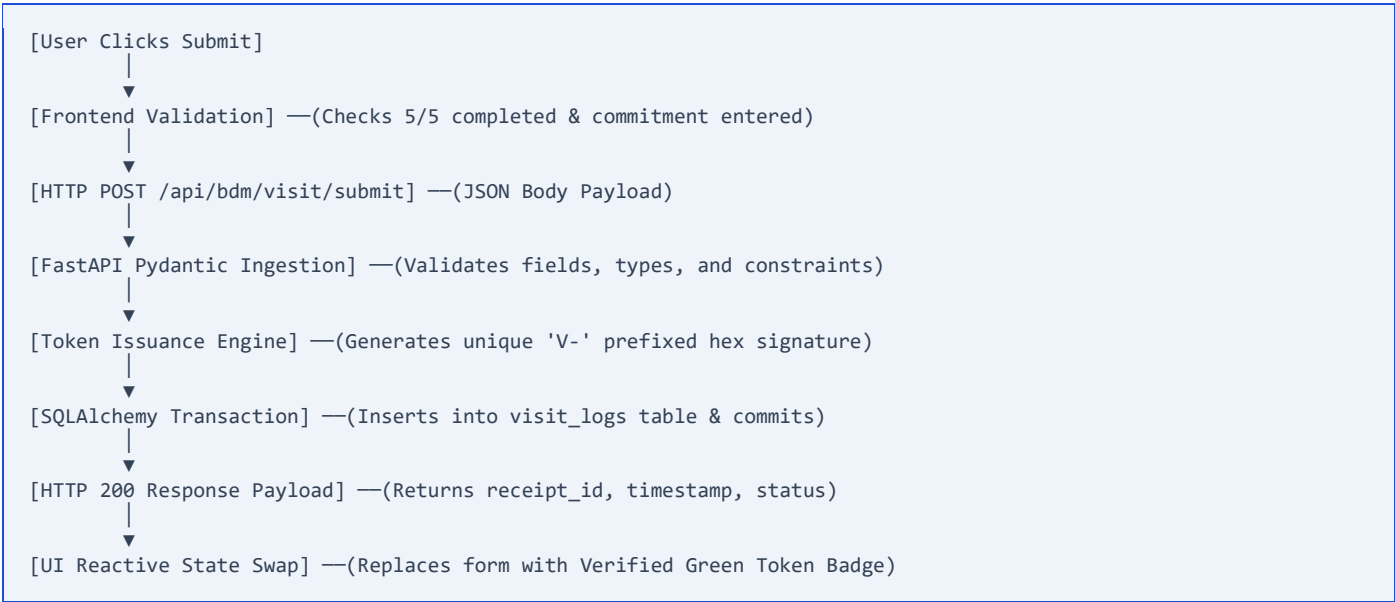
2. End-to-End Data Flow Pipeline

- 1. Territory Hydration:** Client selects BDM code (`BDM002 - Madurai`) → triggers `GET /api/bdm/beat/{bdm_code}`.
- 2. Dynamic Risk Sorting:** Backend queries all outlets in that territory, executes date-differential math against the last billing timestamp, and bubbles dormant accounts (`>45 days` inactive) to index `0..N`.
- 3. Instant Search & Filter:** The client stores the prioritized beat in memory, allowing sub-millisecond filtering across Store Name, Owner Name, and Outlet Code (`OAXXXX`) without extra network round-trips.

4. Dossier Audit Inspection: User selects an outlet → React mounts the channel-specific 5-point action items and populates 6-month run-rate graphs.

3. Deep Dive: "Complete & Verify Visit" Submit Lifecycle

When the BDM fills out the checklist, enters the proprietor's re-order commitment, and clicks "Complete & Verify Visit", the system executes the following atomic sequence:



Channel-Tier Checklists (Core Business Logic)

Tier-Adaptive 5-Point Operational Audit Framework

COLUMN 1

General Trade (GT)
Target Profile: Multi-brand independent dealers, neighborhood mobile shops, high working-capital sensitivity.

1. Shelf Visibility & POSM Placement

Operational Action: Verify that brand standees, lighted headers, acrylic pedestals, and dummy units occupy front-counter, eye-level shelf space. Ensure aggressive multi-brand competitors have not pushed inventory behind counter pillars or cluttered display glass.

2. Overdue COD & Credit Settlement

Operational Action: Review outstanding invoice aging, collect pending Cash-On-Delivery (COD) dues, and clear credit ledger balances with the proprietor. Settling past dues unblocks distributor credit limits so new orders can be invoiced immediately.

3. Margin & Scheme Dispute Reconciliation

Operational Action: Resolve pending trade schemes, backend quarterly cashback calculations, and festival credit-note disputes. Unresolved payout friction is the leading cause of dealers switching walk-in customers to competing brands with instant margins.

4. Fast-Moving SKU Stockout Assessment

Operational Action: Physically inspect backroom and display stock for base storage tiers and fast-moving color variants. Identify zero-inventory SKUs that cause immediate walk-out lost sales.

5. Replenishment Booking & Owner Commitment

Operational Action: Convert stockout gaps into an on-the-spot replenishment order. Log the proprietor's confirmed unit quantity and delivery date directly into the field engine.

COLUMN 2

Mobile Specialist
Target Profile: High-footfall multi-brand smartphone specialist counters with strong ecosystem cross-sell potential.

1. Evaluate Apple Original Accessory Attach Rate

Objective: Maximize revenue per customer transaction and build ecosystem stickiness.

Why It Matters: Smartphones carry narrow hardware margins; the real profitability comes from ecosystem attachments (20W USB-C power adapters, MagSafe cases, AirPods, and AppleCare+).

Audit Action:

- Check billing POS data for attach ratios (e.g., target: ≥ 1.5 accessories per iPhone).
- Identify cross-selling gaps and ensure genuine accessories are displayed right beside hero devices at checkout.

2. Verify Active Consumer Financing & EMI Schemes

Objective: Lower the upfront purchase barrier for price-conscious upgrade buyers.

Why It Matters: Over 60–70% of premium smartphone sales in specialist stores rely on consumer financing (0% interest, no-cost EMI, and instant bank cashbacks).

Audit Action:

- Test POS credit card machines to confirm that bank tie-up cashbacks (e.g., HDFC, ICICI, SBI) and paperless instant loan providers (e.g., Bajaj Finserv, TVS Credit) are functional.
- Verify that promotional EMI table tents and counter acrylics are visible on sales desks.

3. Review Trade-In / Buyback Stock Aging

Objective: Prevent store working capital from being trapped in rapidly depreciating used devices.

Why It Matters: Specialist stores offer instant exchange discounts on old smartphones to close iPhone sales. If those old phones sit in the backroom instead of being dispatched to trade-in partners (e.g., Cashify, Servify), the retailer loses cash flow.

Audit Action:

- Audit backroom storage for collected exchange devices.
- Ensure devices are evaluated, packed, and handed over to logistics partners within 48–72 hours to prevent aging and valuation loss.

4. Review Counter Sales Rep Commissions

Objective: Keep Floor Sales Reps (FSRs) motivated to actively pitch and recommend Apple over competing flagships.

Why It Matters: Floor reps are on the front line. If competitor brands offer higher spot incentives or if official incentive payouts are delayed, reps will switch walk-in customers to rival Android flagships.

Audit Action:

- Review current month sales targets and confirm that payout/incentive slabs are transparent and credited on time.
- Address rep queries on active target bonuses and reward programs.

5. Place Stock Replenishment Order

Objective: Prevent weekend stockouts across high-velocity configurations.

Why It Matters: Mobile Specialists experience high footfall on weekends. Running out of standard storage sizes (128GB/256GB) or trending colors results in immediate lost sales to neighboring stores.

Audit Action:

- Analyze 30-day sales run rates against current backroom stock.
- Secure a binding replenishment order from the store manager/proprietor and confirm the distributor's dispatch date.

COLUMN 3**Apple Premium Reseller (APR)**

The 5-Point Action Checklist for Apple Premium Resellers (APRs) focuses on brand standards, live device health, sales target pacing, credit limit maintenance, and weekly inventory pipeline management. Because APRs are high-volume, mono-brand flagship environments, maintaining pristine customer touchpoints and high inventory throughput is critical.

1. Check Apple Hero Display & Stock Availability

Objective: Ensure flagship hardware visibility aligns with global brand guidelines and meets high walk-in demand.

Operational Action:

- Inspect the central Hero Display tables for correct device positioning, clean security tethers, and pristine screen surfaces.
- Audit backroom stock against hero SKUs (e.g., base storage tiers, trending Pro finish colors) to prevent lost walk-in sales.

2. Verify Working Condition of Demo Units

Objective: Deliver a seamless interactive customer experience across all product families.

Operational Action:

- Test active demo units (iPhones, iPads, MacBooks, Apple Watches) to confirm they are running current official Demo Loop software.
- Check battery health, active power charging under tables, sample media availability (Apple Music playlists, Arcade demos), and screen responsiveness.

3. Review MoM Volume Gap vs. Monthly Target

Objective: Track run-rate pacing to identify revenue shortfalls early in the billing cycle.

Operational Action:

- Compare Month-over-Month (MoM) sell-out performance against the assigned monthly volume and revenue quotas.
- Identify underperforming product categories (e.g., Mac vs. iPhone attach) and formulate tactical weekend drive initiatives with store leadership.

4. Reconcile Outstanding Credit Balances

Objective: Keep distributor credit lines open for seamless, large-ticket weekly stock dispatches.

Operational Action:

- Review open invoices, aging credit balances, and pending credit-note settlements with the store accounts manager.
- Ensure cleared balances prevent billing holds or credit-ceiling blocks from the primary regional distributor.

5. Lock Next Week's Inventory Order Commit

Objective: Secure confirmed stock allocation ahead of peak weekend footfall cycles.

Operational Action:

- Finalize replenishment quantities across hardware, fast-moving original accessories (20W adapters, MagSafe cases, AirPods), and AppleCare+ packs.
- Secure a firm commitment from the store manager and log the order into the field platform for priority distributor dispatch.

Agreed Retailer Action & Next Order Commit

Agreed Retailer Action

Meaning: The specific corrective operational steps the store proprietor or manager promises to take immediately following the audit.

Key Focus Areas: Clearing overdue credit balances, shifting display placement to prime eye-level shelves, repairing demo units, or activating bank EMI promotional standees.

Next Order Commit

Meaning: The binding commercial order quantity and SKU mix the retailer commits to purchase for the next billing cycle.

Key Focus Areas: Base vs. Pro model allocation, high-demand storage sizes, fast-moving color variants, and ecosystem accessory bundles.

Target Fulfillment / Dispatch Timeline

Meaning: The agreed calendar date or delivery window by which the distributor must bill and dispatch the committed stock to the counter.

Engineering Challenge & Resolution

The Problem: SQLite Multi-Threading Concurrency Violation in FastAPI

Error Log

```
sqlite3.ProgrammingError: SQLite objects created in a thread can only be used in that same thread.
```

Root Cause & Fix Explanation

- **Root Cause:** FastAPI dispatches concurrent requests across an async worker thread pool, so a database connection opened on one worker thread was being reused on a different thread for a later request — something the default SQLite driver forbids.
- **The Fix:** Configured SQLAlchemy connection pooling with `connect_args={"check_same_thread": False}` and moved to dynamic, per-request session injection via a `get_db()`.

Verification & Production Roadmap

Testing Rigor, Live Verification & Forward Roadmap

Quality Assurance

- **Automated Test Coverage:** Backend API contract, dormancy-sorting, and receipt-issuance logic are covered by an automated Pytest suite, run via `pytest tests/test_api.py -v`, validating every route against expected status codes and Pydantic schemas.
- **Live Cryptographic Verification:** Each completed visit issues a live, uniquely generated receipt token in the `V-XXXXXX` format, giving distributors and BDMs an on-the-spot, tamper-evident proof of commitment rather than an unverifiable paper slip.

Future Enhancements

Offline-First PWA

Service Workers + IndexedDB caching so beat lists and audit forms stay usable in dead-zone store basements with no live connectivity.

GPS Route Optimization

Optimal daily travel paths across a BDM's assigned outlets via the Mapbox/Google Maps Directions API, cutting windshield time between visits.

WhatsApp Automation

Instant PDF/text receipt dispatch straight to the store proprietor's WhatsApp the moment a visit is verified, replacing manual follow-up.

Distributor ERP Webhooks

Direct order push from confirmed replenishment commitments into regional Tally/SAP backends, removing manual re-entry at the distributor.